Channel-Resolved Conditional Partition Frontier

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Abstract

We consolidate the exact partition-ratio programme after the Haar-entropy, coherent-optimizer, and overlap-correct charging stages. The coherent continuous channel now has a strict conditional budget, while the incoherent channel has a quantified charge–entropy deficit. Center, cut/orbit, and topology channels remain separated so that restricted results are not promoted to the complete $SU(2)$ measure.

1 The exact target

For retained boundary state s, channel c, positive weight V, good reference partition $Z_{G,\mathrm{ref}}$, and charged bad numerator $B_{V,c}$, the open physical target remains

$$\sup_{\Lambda,s,c} \frac{B_{V,c}(s)}{Z_{G,\mathrm{ref}}(s)V(s)} \le e^{-F_*}, \qquad F_* > 6.71292101139.$$

The supremum includes all finite volumes, positive-mass retained states, adaptive histories, and defect channels. The full normalizer cannot be replaced by a pointwise action minimum.

2 Channel ledger

Channel	Strongest result	Physical status
Center	Exact $GF(2)$ five-plaquette charge and restricted contour	Relative activity under the complete continuous measure is open.
Continuous coherent	Unique Haar optimizer and strict conditional $K + R$ budget	Closed only under the common-η collar hypothesis.
Continuous incoherent	Overlap-correct multi-witness charge and connected entropy	Coherent and incoherent feasible η regions are disjoint for current constants.
Cut/orbit	Two-center atlas and exact relative tree-gauge disintegration	Retained-state weighted numerator and transition normalizer are open.
Topology	Scope obstruction identified	Present open local block carries no topology label.

Theorem 1 (Conservative frontier statement)**.** *The package proves two pieces of restricted or conditional progress: the exact center-sector contour and the coherent-collar conditional budget. It proves zero complete physical channel estimates uniform in boundary state and volume. Consequently the exact physical partition ratio, one-scale basin entry, continuum construction, and mass gap remain open.*

Proof. Each row of the executable ledger carries Boolean fields for normalized partition control, boundary-state uniformity, volume uniformity, and physical closure. The center row lacks transfer to the continuous measure; the coherent row retains a hypothesis; the incoherent row fails the common-parameter test; the cut/orbit row lacks a numerator estimate; and the topology row lacks a retained representation. Hence no row meets all physical quantifiers. □

3 The repair frontier

Let η_{coh} be the largest coherence radius accepted at the maximum nonlinear scale. For charge amplification A and effective connected degree D_{eff}, the rooted size-$k_0 = 2$ free energy is

$$2A\frac{\eta_{\mathrm{coh}}^2}{4t} - 2\alpha - 2\log D_{\mathrm{eff}} + \log\Bigl(1 - e^{\alpha} D_{\mathrm{eff}}^2 e^{-A\eta_{\mathrm{coh}}^2/(4t)}\Bigr).$$

The accompanying calibration solves its level set at $F_* = 6.71292101139$. At unit amplification the effective degree must fall to the reported value near 2.2054, far below the present coarse count 18. At degree 18, amplification must reach about 1.55295. Intermediate certified combinations form the preferred next target.

4 Next mathematical target

Construct a gauge-covariant canonical witness polymer that:

1. assigns each incoherent collar to retained, sector-aware witnesses;
2. proves exact overlap and connected-counting constants landing on the repair frontier;
3. compares its weighted numerator to the same retained-state good partition, uniformly over adaptive histories and finite volumes;
4. handles chart transitions and carries an explicit topology label.

Remark 1. *Only after these channel estimates close can the programme claim one-scale nonlinear basin entry. RG iteration, cutoff removal, Osterwalder–Schrader reconstruction, and a positive Hamiltonian gap are later independent nodes.*